Concept 1 | Binomial and Multinomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Binomial

$$\binom{n}{r}a^r b^{n-r}$$

Multinomial

$$\binom{n}{r}\binom{n-r}{s}a^r b^s c^{n-r-s}$$

Exponent check

$$r+(n-r)=n$$

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Worked Solutions **Binomial and Multinomial Theorems**

Q001 **Challenge**

For $(2x-3)^5$, find the coefficient of x^3 and the sum of all coefficients without writing the full expansion:

Solution

1. Use the binomial term rule: $q = 3$ and $r = 2$, so the coefficient is 5C2 times 72.
2. For the sum of coefficients, evaluate the expression at $x = 1$: -1.

Correct option: A

Final answer

Coefficient = 720; sum = -1.

Worked Solutions **Binomial and Multinomial Theorems**

Q002 **Challenge**

Find the coefficient of x^2y^3z in $(x+y+z)^6$, then find the coefficient of x^2yz^3 in $(x-2y+z)^6$:

Solution

1. The first exponents are (2,3,1), so use 6!/(2!3!1!) = 60.
2. The second exponents are (2,1,3); the factor (-2) is raised to the first power of y, giving -120.

Correct option: A

Final answer

First coefficient = 60; second coefficient = -120.

Worked Solutions **Binomial and Multinomial Theorems**

Q003 **Challenge**

For $(x^2+x+1)^4$, find the greatest coefficient and verify it by identifying the coefficient of x^4:

Solution

1. Expand and collect like powers: the central coefficient is 19.
2. The coefficient of x^4 is also 19; it is therefore the greatest.

Correct option: A

Final answer

Greatest coefficient = 19.

Worked Solutions **Binomial and Multinomial Theorems**

Q004 Written

Expand $(x+2)^5$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+2)^5 = x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$.

Final answer

$x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$

Q005 MCQ

Expand $(3x-2)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x-2)^4 = 81x^4 - 216x^3 + 216x^2 - 96x + 16$.

Correct option: D

Final answer

$81x^4 - 216x^3 + 216x^2 - 96x + 16$

Worked Solutions **Binomial and Multinomial Theorems**

Q006 Written

Expand $(x+2)^6$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+2)^6 = x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$.

Final answer

$x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$

Q007 MCQ

Expand $(x+1)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1$.

Correct option: B

Final answer

$x^4 + 4x^3 + 6x^2 + 4x + 1$

Worked Solutions **Binomial and Multinomial Theorems**

Q008 MCQ

Expand $(a+b)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$.

Correct option: C

Final answer

$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

Q009 Written

Expand $(a-2b)^4$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a-2b)^4 = a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$.

Final answer

$a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$

Worked Solutions **Binomial and Multinomial Theorems**

Q010 MCQ

Expand $(x-2)^5$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x-2)^5 = x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$.

Correct option: A

Final answer

$x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$

Q011 Written

Expand $(3x+2y)^5$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x+2y)^5 = 243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$.

Final answer

$243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$

Worked Solutions **Binomial and Multinomial Theorems**

Q012 Written

Expand $(2a^2 - 1)^5$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a^2 - 1)^5 = 32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$.

Final answer

$32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$

Q013 Written

Expand $(x - 3)^6$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 3)^6 = x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$.

Final answer

$x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$

Worked Solutions **Binomial and Multinomial Theorems**

Q014 **Written**

Expand $(2x - 3y)^6$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x - 3y)^6 = 64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$.

Final answer

$64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$

Q015 **Written**

Expand $\left(x + \frac{1}{3}\right)^6$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $\left(x + \frac{1}{3}\right)^6 = x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$.

Final answer

$$x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$$

Worked Solutions **Binomial and Multinomial Theorems**

Q016 **MCQ**

Expand $(x+3)^3$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+3)^3 = x^3 + 9x^2 + 27x + 27$.

Correct option: B

Final answer

$x^3 + 9x^2 + 27x + 27$

Worked Solutions **Binomial and Multinomial Theorems**

Q017 **MCQ**

Expand $(a+2b)^5$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a+2b)^5$=$a^5+10a^4b+40a^3b^2+80a^2b^3+80ab^4+32b^5$.

Correct option: D

Final answer

$a^5+10a^4b+40a^3b^2+80a^2b^3+80ab^4+32b^5$

Worked Solutions **Binomial and Multinomial Theorems**

Q018 MCQ

Expand $(x+2)^2$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x+2)^2 = x^2 + 4x + 4$.

Correct option: D

Final answer

$x^2 + 4x + 4$

Q019 MCQ

Expand $(a-b)^3$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$.

Correct option: A

Final answer

$a^3 - 3a^2b + 3ab^2 - b^3$

Worked Solutions **Binomial and Multinomial Theorems**

Q020 MCQ

Expand $(2x+1)^5$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x+1)^5$=$32x^5+80x^4+80x^3+40x^2+10x+1$.

Correct option: B

Final answer

$32x^5+80x^4+80x^3+40x^2+10x+1$

Q021 Written

Expand $(3a-2b)^5$ using the Binomial Theorem.

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3a-2b)^5$=$243a^5-810a^4b+1080a^3b^2-720a^2b^3+240ab^4-32b^5$.

Final answer

$243a^5-810a^4b+1080a^3b^2-720a^2b^3+240ab^4-32b^5$

Worked Solutions **Binomial and Multinomial Theorems**

Q022 MCQ

Expand $(2x-1)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x-1)^4$=$16x^4-32x^3+24x^2-8x+1$.

Correct option: C

Final answer

$16x^4-32x^3+24x^2-8x+1$

Q023 MCQ

Expand $(x-3)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x-3)^4$=$x^4-12x^3+54x^2-108x+81$.

Correct option: B

Final answer

$x^4-12x^3+54x^2-108x+81$

Worked Solutions **Binomial and Multinomial Theorems**

Q024 MCQ

Expand $(2a+b)^3$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a+b)^3$=$8a^3+12a^2b+6ab^2+b^3$.

Correct option: C

Final answer

$8a^3+12a^2b+6ab^2+b^3$

Q025 MCQ

Expand $\left(2a^2+3\right)^4$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $\left(2a^2+3\right)^4$=$16a^8+96a^6+216a^4+216a^2+81$.

Correct option: C

Final answer

$16a^8+96a^6+216a^4+216a^2+81$

Worked Solutions **Binomial and Multinomial Theorems**

Q026 MCQ

Expand $\left(3a^2-2\right)^3$ using the Binomial Theorem:

Solution

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $\left(3a^2-2\right)^3$=$27a^6-54a^4+36a^2-8$.

Correct option: B

Final answer

$27a^6-54a^4+36a^2-8$

Q027 Written

Find the coefficient of a^3b^2 in the expansion of $(a+b)^5$.

Solution

1. The target power uses $q=3$ copies of the first term, so $r=n-q=2$.
2. Use 5C2 times the first factor to power q and the second factor to power r.
3. The required coefficient is 10.

Final answer

10

Worked Solutions **Binomial and Multinomial Theorems**

Q028 **Written**

Find the coefficient of x^4y^2 in the expansion of $(x-2y)^6$.

Solution

1. The target power uses $q=4$ copies of the first term, so $r=n-q=2$.
2. Use 6C2 times the first factor to power q and the second factor to power r.
3. The required coefficient is 60.

Final answer

60

Q029 **Written**

Find the coefficient of x^4 in the expansion of $\left(2x^2-3\right)^7$.

Solution

1. The target power uses $q=2$ copies of the first term, so $r=n-q=5$.
2. Use 7C5 times the first factor to power q and the second factor to power r.
3. The required coefficient is -20412.

Final answer

-20412

Worked Solutions **Binomial and Multinomial Theorems**

Q030 Written

Find the coefficient of x^6y in the expansion of $(2x-3y)^7$.

Solution

1. The target power uses $q=6$ copies of the first term, so $r=n-q=1$.
2. Use 7C1 times the first factor to power q and the second factor to power r.
3. The required coefficient is -1344.

Final answer

-1344

Q031 MCQ

Find the coefficient of x^2 in the expansion of $(3x-2)^5$:

Solution

1. The target power uses $q=2$ copies of the first term, so $r=n-q=3$.
2. Use 5C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is -720.

Correct option: B

Final answer

-720

Worked Solutions **Binomial and Multinomial Theorems**

Q032 Written

Find the coefficient of a^6 in the expansion of $\left(3a^2 - 2\right)^8$.

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 5$.
2. Use 8C5 times the first factor to power q and the second factor to power r.
3. The required coefficient is -48384.

Final answer

-48384

Q033 MCQ

Find the coefficient of x^4y^4 in the expansion of $(2x - y)^8$:

Solution

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 4$.
2. Use 8C4 times the first factor to power q and the second factor to power r.
3. The required coefficient is 1120.

Correct option: A

Final answer

1120

Worked Solutions **Binomial and Multinomial Theorems**

Q034 MCQ

Find the coefficient of ab^3 in the expansion of $(3a+b)^4$:

Solution

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use 4C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is 12.

Correct option: B

Final answer

12

Q035 MCQ

Find the coefficient of x^3y^2 in the expansion of $(x-4y)^5$:

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C2 times the first factor to power q and the second factor to power r.
3. The required coefficient is 160.

Correct option: C

Final answer

160

Worked Solutions **Binomial and Multinomial Theorems**

Q036 MCQ

Find the coefficient of x^4 in the expansion of $(3x + 1)^5$:

Solution

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 1$.
2. Use 5C1 times the first factor to power q and the second factor to power r.
3. The required coefficient is 405.

Correct option: D

Final answer

405

Q037 Written

Find the coefficient of a in the expansion of $(2a - 3)^4$.

Solution

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use 4C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is -216.

Final answer

-216

Worked Solutions **Binomial and Multinomial Theorems**

Q038 Written

Find the coefficient of x^3 in the expansion of $(3x+2)^5$.

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C2 times the first factor to power q and the second factor to power r.
3. The required coefficient is 1080.

Final answer

1080

Q039 Written

Find the coefficient of x^6 in the expansion of $\left(2x^2-1\right)^6$.

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use 6C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is -160.

Final answer

-160

Worked Solutions **Binomial and Multinomial Theorems**

Q040 MCQ

Find the coefficient of x^4 in the expansion of $\left(3x^2-1\right)^5$:

Solution

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 3$.
2. Use 5C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is -90.

Correct option: D

Final answer

-90

Q041 Written

Find the coefficient of x^6 in the expansion of $\left(3x^2+2\right)^6$.

Solution

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use 6C3 times the first factor to power q and the second factor to power r.
3. The required coefficient is 4320.

Final answer

4320

Worked Solutions **Binomial and Multinomial Theorems**

Q042 Written

If $(x+1)^5 = a_0 + a_x + \ldots$, find the sum of its coefficients.

Solution

1. Put $x = 1$; the sum of the coefficients is the value of the expansion at $x = 1$.
2. Therefore the required value is 32.

Final answer

32

Q043 MCQ

Find the coefficient of xyz in the expansion of $(x+y+z)^3$.:

Solution

1. The target powers are $p = 1,\ q = 1,\ r = 1$, and $p + q + r = 3$.
2. Coefficient = $\dfrac{3!}{1!\ \times\ 1!\ \times\ 1!} \times (1)^1 \times (1)^1 \times (1)^1 = 6$.
3. Therefore, the required coefficient is 6.

Correct option: A

Final answer

6

Worked Solutions **Binomial and Multinomial Theorems**

Q044 **MCQ**

Find the coefficient of a^4b^3c in the expansion of $(a+b+c)^8$:

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p! \times q! \times r!}$ = 280 on the signed factors.
3. The required coefficient is 280.

Correct option: C

Final answer

280

Q045 **MCQ**

Find the coefficient of a^3bc^2 in the expansion of $(a+b+c)^6$:

Solution

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\dfrac{n!}{p! \times q! \times r!}$ = 60 on the signed factors.
3. The required coefficient is 60.

Correct option: A

Final answer

60

Worked Solutions **Binomial and Multinomial Theorems**

Q046 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a+b+c)^6$:

Solution

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 90.

Correct option: B

Final answer

90

Q047 MCQ

Find the coefficient of xy^5z in the expansion of $(x+y+z)^7$:

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is 42.

Correct option: A

Final answer

42

Worked Solutions **Binomial and Multinomial Theorems**

Q048 Written

Find the coefficient of x^4y^2z in the expansion of $(x+y+z)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!}$ = 105 on the signed factors.
3. The required coefficient is 105.

Final answer

105

Q049 Written

Find the coefficient of $x^2y^3z^2$ in the expansion of $(x+y+z)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!}$ = 210 on the signed factors.
3. The required coefficient is 210.

Final answer

210

Worked Solutions **Binomial and Multinomial Theorems**

Q050 **Written**

Find the coefficient of x^6yz^2 in the expansion of $(x+y+z)^9$.

Solution

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\dfrac{n!}{p! \times q! \times r!}$ = 252 on the signed factors.
3. The required coefficient is 252.

Final answer

252

Q051 **Written**

Find the coefficient of $x^3y^3z^3$ in the expansion of $(x+y+z)^9$.

Solution

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\dfrac{n!}{p! \times q! \times r!}$ = 1680 on the signed factors.
3. The required coefficient is 1680.

Final answer

1680

Worked Solutions **Binomial and Multinomial Theorems**

Q052 Written

Find the coefficient of $x^2y^4z^3$ in the expansion of $(x+y+z)^9$.

Solution

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\dfrac{n!}{p!\ \times\ q!\ \times\ r!}$ = 1260 on the signed factors.
3. The required coefficient is 1260.

Final answer

1260

Q053 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a-2b+3c)^8$:

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p!\ \times\ q!\ \times\ r!}$ = 280 on the signed factors.
3. The required coefficient is −6720.

Correct option: D

Final answer

−6720

Worked Solutions **Binomial and Multinomial Theorems**

Q054 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + 3b - 2c)^6$:

Solution

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 3240.

Correct option: C

Final answer

3240

Q055 MCQ

Find the coefficient of ab^2c^3 in the expansion of $(a + 3b - 2c)^6$:

Solution

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 60$ on the signed factors.
3. The required coefficient is -4320.

Correct option: D

Final answer

-4320

Worked Solutions **Binomial and Multinomial Theorems**

Q056 **Written**

Find the coefficient of $a^2b^3c^2$ in the expansion of $(a+2b-3c)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 210$ on the signed factors.
3. The required coefficient is 15120.

Final answer

15120

Q057 **Written**

Find the coefficient of abc^5 in the expansion of $(a+2b-3c)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is -20412.

Final answer

-20412

Worked Solutions **Binomial and Multinomial Theorems**

Q058 Written

Find the coefficient of a^2b^4c in the expansion of $(a + 2b - 3c)^7$.

Solution

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 105$ on the signed factors.
3. The required coefficient is -5040.

Final answer

-5040

Q059 Written

Find the coefficient of $a^2b^2c^4$ in the expansion of $(2a - b + 3c)^8$.

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 420$ on the signed factors.
3. The required coefficient is 136080.

Final answer

136080

Worked Solutions **Binomial and Multinomial Theorems**

Q060 Written

Find the coefficient of a^4bc^3 in the expansion of $(2a - b + 3c)^8$.

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is -120960.

Final answer

-120960

Q061 Written

Find the coefficient of $a^3b^3c^2$ in the expansion of $(2a - b + 3c)^8$.

Solution

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\dfrac{n!}{p! \times q! \times r!} = 560$ on the signed factors.
3. The required coefficient is -40320.

Final answer

-40320

Worked Solutions **Binomial and Multinomial Theorems**

Q062 Written

Find the greatest coefficient in the expansion of $\left(x^2+x+1\right)^4$.

Solution

1. Expand and collect like powers: $x^8+4x^7+10x^6+16x^5+19x^4+16x^3+10x^2+4x+1$.
2. The greatest coefficient is 19.

Final answer

19

Quiz MCQ 1 Concept Quiz · MCQ

The coefficient of x^4 in $(2x-3)^5$ is:

Solution

1. $q=4$ and $r=1$, so the coefficient is -240.

Correct option: A

Final answer

-240

Worked Solutions **Binomial and Multinomial Theorems**

Quiz MCQ 2 **Concept Quiz · MCQ**

The coefficient of $x^2y^2z^2$ in $(x+y+z)^6$ is:

Solution

1. Use 6!/(2!2!2!).

Correct option: C

Final answer

90

Quiz MCQ 3 **Concept Quiz · MCQ**

The coefficient of x^6 in $\left(3x^2-2\right)^6$ is:

Solution

1. Choose the first factor three times.

Correct option: A

Final answer

-4320

Worked Solutions **Binomial and Multinomial Theorems**

Quiz MCQ 4 Concept Quiz · MCQ

The coefficient of x^5 in $\left(2x^2+1\right)^4$ is:

Solution

1. All powers of x are even, so the coefficient is zero.

Correct option: A

Final answer

0

Quiz WRITTEN 5 Concept Quiz · Written

Expand $(2x-1)^4$ fully.

Solution

1. The expansion is $16x^4-32x^3+24x^2-8x+1$.

Final answer

$16x^4-32x^3+24x^2-8x+1$

Worked Solutions **Binomial and Multinomial Theorems**

Quiz WRITTEN 6 Concept Quiz · Written

Find the coefficient of x^4 in $\left(3x^2+2\right)^5$. Show the exponent check.

Solution

1. The first-term exponent is $q = 2$ and $r = 3$, giving 5C3 × 9 × 8=720.

Final answer

720

Quiz WRITTEN 7 Concept Quiz · Written

Find the coefficient of x^2y^3z in $(x - 2y + z)^6$.

Solution

1. Use the multinomial coefficient and the signed factor; the result is −480.

Final answer

−480

Worked Solutions **Binomial and Multinomial Theorems**

Quiz WRITTEN 8 **Concept Quiz · Written**

Find the greatest coefficient in $(x^2 + x + 1)^4$. Show enough work to justify the maximum.

Solution

1. Expand: $x^8 + 4x^7 + 10x^6 + 16x^5 + 19x^4 + 16x^3 + 10x^2 + 4x + 1$; the greatest coefficient is 19.

Final answer

19

Answer key

Use the question number shown on each page.

Q001 A

Q002 A

Q003 A

Q004 $x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$

Q005 D

Q006 $x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$

Q007 B

Q008 C

Q009 $a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$

Q010 A

Q011 $243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$

Q012 $32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$

Q013 $x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$

Q014 $64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$

Q015 $x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$

Q016 B

Q017 D

Q018 D

Q019 A

Q020 B

Q021 $243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$

Q022 C

Q023 B

Q024 C

Q025 C

Q026 B

Q027 10

Q028 60

Q029 -20412

Q030 -1344

Q031 B

Q032 -48384

Q033 A

Q034 B

Q035 C

Q036 D

Q037 -216

Q038 1080

Q039 -160

Q040 D

Q041 4320

Q042 32

Q043 A

Q044 C

Q045 A

Q046 B

Q047 A

Q048 105

Q049 210

Q050 252

Q051 1680

Q052 1260

Q053 D

Q054 C

Q055 D

Q056 15120

Q057 -20412

Q058 -5040

Q059 136080

Q060 -120960

Q061 -40320

Q062 19

Quiz MCQ 1 A

Quiz MCQ 2 C

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $16x^4 - 32x^3 + 24x^2 - 8x + 1$

Quiz WRITTEN 6 720

Quiz WRITTEN 7 -480

Quiz WRITTEN 8 19